

Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C

Chapter 6 Flow Control in Assembly Exercises ANS

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Pseudocode to Assembly

- ▶ Write assembly program for pseudocode, one version without Conditional Execution instructions, one version with Conditional Execution.

Pseudocode	Assembly Program
if (r0 != r1) r2 = r2 + r2	

Pseudocode	Assembly Program w/Conditional Execution
if (r0 != r1) r2 = r2 + r2	

Pseudocode to Assembly ANS

- ▶ Write assembly program for pseudocode, one version without Conditional Execution instructions, one version with Conditional Execution.
- ▶ if ($r0 \neq r1$) $r2 = r2 + r2$

Pseudocode	Assembly Program
if ($r0 \neq r1$) $r2 = r2 + r2$	CMP $r0, r1$ BEQ skip ADD $r2, r2, r2$ skip:

Pseudocode	Assembly Program w/Conditional Execution
if ($r0 \neq r1$) $r2 = r2 + r2$	CMP $r0, r1$ ADDNE $r2, r2, r2$

Pseudocode to Assembly

- ▶ Write assembly program for pseudocode.

Pseudocode	Assembly Program 1
<code>r1 = (r0 >> 4) & 15</code>	

Pseudocode to Assembly ANS

- ▶ Write assembly program for pseudocode.

Pseudocode	Assembly Program 1
$r1 = (r0 \gg 4) \& 15$	MOV r1, r0, LSR #4 AND r1, r1, 0x0F %AND r1, r1, #15 is also OK

Pseudocode	Assembly Program 2
$r1 = (r0 \gg 4) \& 15$	AND r1, r0, #0xF0 MOV r1, r1, LSR #4

- ▶ Assembly Program 1: It computes $r1 = (r0 \gg 4) \& 15$ shifting it down by 4 bits, then masking the low nibble (4 bits) with 0x0F (or 0xF).
- ▶ Assembly Program 2: It computes $r1 = (r0 \gg 4) \& 15$ by first masking the high nibble with 0xF0 and then shifting it down by 4 bits.
 - ▶ AND r1, r0, #0xF0 keeps only bits [7:4] of r0, clearing all others.
 - ▶ MOV r1, r1, LSR #4 performs a logical right shift by 4 on the masked value, moving those bits into positions [3:0].

Assembly Program Understanding

- ▶ What is r0 equal to after running this program:
 - ▶ MOV r0, #10
 - ▶ MOV r1, #7
 - ▶ MOV r2, #5
 - ▶ CMP r1, r2
 - ▶ ADDGT r0, r0, #100

Assembly Program Understanding ANS

- ▶ What is r0 equal to after running this program:
 - ▶ MOV r0,#10 ; Load immediate value 10 into r0
 - ▶ MOV r1,#7 ; Load immediate value 7 into r1
 - ▶ MOV r2,#5 ; Load immediate value 5 into r2
 - ▶ CMP r1,r2 ; Compare r1 and r2 (compute r1-r2, set flags)
 - ▶ ADDGT r0,r0,100 ; Add 100 to r0 only if r1 > r2 (Greater Than)
- ▶ This code sets r0 to 10, compares 7 and 5. Since $7 > 5$, adds 100 to the base value of 10 to get 110
- ▶ What about:
 - ▶ MOV r0,#10
 - ▶ MOV r1,#7
 - ▶ MOV r2,#5
 - ▶ SUBS r1,r2
 - ▶ ADDGE r0,r0,#100
- ▶ Here $r0 = 110$, since SUBS sets flags and ADDGE is executed

C to Assembly

- ▶ `save[]` is an array of 32-bit integers. Assume that `i` and `k` correspond to registers `r1` and `r2`, and the base of the array `save` is in `r0`. Write assembly code corresponding to this C code.

C code	Assembly Program
<code>while (save[i] == k)</code> <code>i += 1;</code>	<code>; r0 = &save[0]</code> (base address of array) <code>; r1 = i</code> (index) <code>; r2 = k</code> (value to compare) <code>...</code>

C to Assembly ANS

- ▶ `save[]` is an array of 32-bit integers. Assume that `i` and `k` correspond to registers `r1` and `r2`, and the base of the array `save` is in `r0`. Write assembly code corresponding to this C code.

C code	Assembly Program
<pre>i = 0; while (save[i] == k) i += 1;</pre>	<pre>; r0 = &save[0] (base address of array) ; r1 = i (index) ; r2 = k (value to compare) loop: LDR r3, [r0, r1, LSL #2] ; r3 = save[i], 32-bit loads with index scaled by 4, so r3 is loaded from mem address r0+4*r1 CMP r3, r2 ; compare save[i] with k BNE done ; exit loop if save[i] != k ADD r1, r1, #1 ; i += 1 B loop ; repeat done:</pre>

C to Assembly

- ▶ Write the equivalent assembly program for this piece of code in C.

C code	Assembly Program
<pre>for (i=0; i<8; i++){ a[i] = b[7-i]; }</pre>	<pre>; Assume r0 = base address of a, r1 = base address of b</pre>

C to Assembly ANS

- ▶ Write the equivalent assembly program for this piece of code in C.

C code	Assembly Program
<pre>for (i=0; i<8; i++){ a[i] = b[7-i]; }</pre>	<pre>; Assume r0 = base address of a, r1 = base address of b MOV r2, #0 ; i = 0 loop_start: CMP r2, #8 BGE loop_end RSB r3, r2, #7 ; r3 = 7 - i LDR r4, [r1, r3, LSL#2] ; Load b[7-i] from mem address r1 + 4*r3 STR r4, [r0, r2, LSL#2] ; Store to a[i] to mem address r0 + 4*r2 ADD r2, r2, #1 ; i++ B loop_start loop_end:</pre>

C to Assembly: What is wrong?

C Program	Assembly Program
<pre>int cnt = 1; while (cnt <= 10) { // loop body cnt++; }</pre>	<pre>MOV r0, #1 ; cnt = 1 loop: CMP r0, #10 ; Compare r0 with 10 while r0 <=10 BEQ end ; If cnt == 10, branch to end ADD r0, r0, #1 ; cnt = cnt + 1 B loop ; Repeat the loop done:</pre>

C Program	Assembly Program
<pre>int cnt = 10; while (cnt != 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUB r0, r0, #1 ; cnt--; sets flags BNE loop ; repeat until cnt == 0 (10 times) done:</pre>

C to Assembly: What is wrong? ANS

C Program	Assembly Program
<pre>int cnt = 1; while (cnt <= 10) { // loop body cnt++; }</pre>	<pre>MOV r0, #1 ; cnt = 1 loop: CMP r0, #11 ; Compare x0 with 10 while x0 <=10 BEQ done ; If cnt == 10, branch to end ADD r0, r0, #1 ; cnt = cnt + 1 B loop ; Repeat the loop done:</pre>

Loop runs for 9 iterations, not 10

C Program	Assembly Program
<pre>int cnt = 10; while (cnt != 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations B check loop: SUBS r0, r0, #1 ; cnt--; sets flags check: BNE loop ; repeat until cnt == 0 (10 times) done:</pre>

SUB should be SUBS to set flags

Add a B check before loop body

(program behavior the same if cnt = r0 = 10 initially, but different if cnt = r0 = 0 initially. c.f. pp 22-25 in Ch6 lecture on while loop and do-while loop.)

C to Assembly

C Program	Assembly Program
<pre>int cnt = 1; while (cnt <= 10) { // loop body cnt++; }</pre>	<pre>MOV r0, #1 ; cnt = 1 loop: ; loop body ADD ??? ; cnt++ CMP ??? BLE ??? ; while (cnt <= 10) continue done:</pre>

C Program	Assembly Program
<pre>int cnt = 10; while (cnt != 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS ??? ; cnt--; sets Z flag if r0=0 BNE ??? ; repeat until cnt == 0 (10 times) done:</pre>

C Program	Assembly Program
<pre>int cnt = 10; while (cnt > 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS ???; cnt--; sets Z flag if r0=0 BGT ??? ; repeat until cnt == 0 (10 times) done:</pre>

C to Assembly ANS

C Program	Assembly Program
<pre>int cnt = 1; while (cnt <= 10) { // loop body cnt++; }</pre>	<pre>MOV r0, #1 ; cnt = 1 loop: ; loop body ADD r0, r0, #1 ; cnt++ CMP r0, #10 BLE loop ; while (cnt <= 10) continue done:</pre>

C Program	Assembly Program
<pre>int cnt = 10; while (cnt != 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS r0, r0, #1 ; cnt--; sets Z flag if r0=0 BNE loop ; repeat until cnt == 0 (10 times) done:</pre>

C Program	Assembly Program
<pre>int cnt = 10; while (cnt > 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS r0, r0, #1 ; cnt--; sets Z flag if r0=0 BGT loop ; repeat until cnt == 0 (10 times) done:</pre>

C to Assembly: What is wrong?

C Program	Assembly Program
<pre>int cnt = 10; while (cnt > 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS r0, r0, #1 ; cnt--; sets Z flag if r0=0 BPL loop ; repeat until cnt == 0 (10 times) done:</pre>

C to Assembly: What is wrong? ANS

C Program	Assembly Program
<pre>int cnt = 10; while (cnt > 0) { // loop body cnt--; }</pre>	<pre>MOV r0, #10 ; remaining iterations loop: ; loop body SUBS r0, r0, #1 ; cnt--; sets Z flag if r0=0 BPL loop ; repeat until cnt == 0 (10 times) done:</pre>

BPL, Branch if PLus (Positive or Zero, N = 0), tests the Negative flag N=0 (i.e., $r0 \geq 0$). Starting from 10 and decrementing, N stays 0 for 10 down to 0, so BPL would still branch at $r0 = 0$. Then SUBS makes $r0 = -1$, which sets $N = 1$, so BPL does not branch and the loop exits. Hence BPL allows the loop to run at 0 and overshoot by one iteration.

What's wrong: BPL tests for ≥ 0 , not $> 0 \rightarrow$ extra iteration.

How to fix: Replace BPL loop with BGT loop.

If the loop variable were unsigned, you'd instead use BHI ("branch if higher") for the same effect.

Quiz: is BPL same as BGE?

ANS: No. PL checks $N == 0$; GE checks $N == V$. In this program BGE and BPL have the same behavior, but in general not, c.f. p. 16 in Ch6 lecture

C to Assembly

C Program	Assembly Program
<pre>int array[200]; int i; for (i = 199; i >= 0; i = i - 1) { array[i] = array[i] * 8; }</pre>	<pre>% R0 = base address of array, R1 = i MOV R0, 0x60000000 ; base address where array resides MOV R1, #199 ; i = 199 ...</pre>

C to Assembly ANS

C Program	Assembly Program
<pre>int array[200]; int i; for (i = 199; i >= 0; i = i - 1) { array[i] = array[i] * 8; }</pre>	<pre>% R0 = base address of array, R1 = i MOV R0, 0x60000000 ; base address where array resides MOV R1, #199 ; i = 199 FOR: LDR R2, [R0, R1, LSL #2] ; R2 = array[i] (i*4 byte offset) LSL R2, R2, #3 ; R2 = R2 * 8 STR R2, [R0, R1, LSL #2] ; array[i] = R2 SUBS R1, R1, #1 ; i = i - 1, set flags BPL FOR ; while (i >= 0) repeat %BGE FOR ; BGE is also OK here</pre>

C to Assembly

C Program	Assembly Program
<pre>//Calculate x such that 2^x=128 int pow = 1; int x = 0; while (pow != 128) { pow = pow * 2; x = x + 1; }</pre>	<pre>; r0 = pow, r1 = x MOV r0, #1 ; pow = 1 MOV r1, #0 ; x = 0 WHILE DONE</pre>

C to Assembly ANS

C Program	Assembly Program
<pre>//Calculate x such that 2^x = 128 int pow = 1; int x = 0; while (pow != 128) { pow = pow * 2; x = x + 1; }</pre>	<pre>; r0 = pow, r1 = x MOV r0, #1 ; pow = 1 MOV r1, #0 ; x = 0 WHILE CMP r0, #128 ; r0-128 BEQ DONE ; if (pow==128) exit loop LSL r0, r0, #1 ; pow = pow * 2 ADD r1, r1, #1 ; x = x + 1 B WHILE ; repeat loop DONE</pre>



C to Assembly ANS

C Program	Assembly Program
<pre>//Calculate x such that 2^x = 128 int pow = 1; int x = 0; while (pow != 128) { pow = pow * 2; x = x + 1; }</pre>	<pre>; r0 = pow, r1 = x MOV r0, #1 ; pow = 1 MOV r1, #0 ; x = 0 WHILE LSL r0, r0, #1 ; pow = pow * 2 ADD r1, r1, #1 ; x = x + 1 CMP r0, #128 ; r0-128 BNE WHILE ; repeat loop DONE</pre>

- ▶ **Edge case:** If r0 starts as 128 (zero-iteration case), this version will execute the body once (doubling to 256 and incrementing r1) before checking — i.e., wrong for that initial condition, c.f. pp 22-25 in Ch6 lecture on while loop and do-while loop.

C to Assembly

C Program	Assembly Program
<pre>int array[200]; int i; for (i = 199; i >= 0; i=i-1) array[i] = array[i] * 8;</pre>	<pre>; R0 = array base address, R1 = i MOV R0, 0x60000000 MOV R1, #199 FOR</pre>

- ▶ For an array of 32-bit ints, each element is 4 bytes, so element i lives at $\text{base} + i * 4$, which is implemented as $\text{base} + (i \ll 2)$ via `LSL #2` in the addressing mode
- ▶ BPL “Branch if PLus” branches when the N (negative) flag $N == 0$, meaning the prior result was positive or zero

C to Assembly ANS

C Program	Assembly Program
<pre>int array[200]; int i; for (i = 199; i >= 0; i = i - 1) array[i] = array[i] * 8;</pre>	<pre>; R0 = array base address, R1 = i MOV R0, 0x60000000 MOV R1, #199 FOR LDR R2, [R0, R1, LSL #2] ;R2 = array(i) LSL R2, R2, #3 ; R2 = R2<<3 = R2*8 STR R2, [R0, R1, LSL #2] ;array(i) = R2 SUBS R1, R1, #1 ; i=i-1 and set flags BPL FOR ; if (i >= 0) repeat loop</pre>

- ▶ For an array of 32-bit ints, each element is 4 bytes, so element i lives at $\text{base} + i*4$, which is implemented as $\text{base} + (i \ll 2)$ via `LSL #2` in the addressing mode
- ▶ `BPL` “Branch if PLus” branches when the `N` (negative) flag `N == 0`, meaning the prior result was positive or zero

C to Assembly

C Program	Assembly Program
<pre>int i; int sum = 0; for (i = 1; i <= 22; i++) sum += i;</pre>	



C to Assembly ANS

C Program	Assembly Program 1
<pre>int i; int sum = 0; for (i = 1; i <= 22; i++) sum += i;</pre>	<pre> mov r1, #0 /* r1 ← 0 */ mov r2, #1 /* r2 ← 1 */ loop: cmp r2, #22 /* compare r2 and 22 */ bgt end /* branch if r2 > 22 to end */ add r1, r1, r2 /* r1 ← r1 + r2 */ add r2, r2, #1 /* r2 ← r2 + 1 */ b loop end:</pre>
C Program	Assembly Program 2
<pre>int i; int sum = 0; for (i = 1; i <= 22; i++) sum += i;</pre>	<pre> mov r1, #0 /* r1 ← 0 */ mov r2, #1 /* r2 ← 1 */ b check_loop /* unconditionally jump at the end of the loop */ loop: add r1, r1, r2 /* r1 ← r1 + r2 */ add r2, r2, #1 /* r2 ← r2 + 1 */ check_loop: cmp r2, #22 /* compare r2 and 22 */ ble loop /* branch if r2 <= 22 to the beginning of the loop */ end:</pre>

C to Assembly ANS

▶ Assembly Program 1:

- ▶ Two branches per iteration in the steady state: a conditional (cmp + bgt) and an unconditional b loop. That increases dynamic branch count.

▶ Assembly Program 2:

- ▶ Only one conditional branch per iteration (the ble), after the body — fewer dynamic branches overall. In steady-state this is usually faster.
- ▶ Better steady-state throughput and fewer branch misprediction opportunities in hot loops (one backward taken conditional is a canonical, well-predicted pattern).

Assembly Programming

- ▶ Write a program that reverses the bits in a register, such that the register containing `d31,d30,d29...d1,d0` now contains `d0,d1,...d29,d30,d31`.

Assembly Programming ANS

▶ ANS:

- ▶ ; Reverse bits in r0, result in r1
- ▶ MOV r1, #0 ; Initialize result
- ▶ MOV r2, #32 ; Bit counter

▶ reverse_loop:

- ▶ CMP r2, #0
- ▶ BEQ reverse_end
- ▶ LSL r1, r1, #1 ; Shift result left
- ▶ AND r3, r0, #1 ; Get LSB of r0
- ▶ ORR r1, r1, r3 ; Add to result
- ▶ LSR r0, r0, #1 ; Shift r0 right
- ▶ SUB r2, r2, #1 ; Decrement counter
- ▶ B reverse_loop

▶ reverse_end:

▶ Example with 8 bits:

▶ Let input r0 = 0b1011_0010

▶ Initialize r1 = 0, r2 = 8 (bit counter), and r3 is the scratch for the extracted bit; the loop body runs 8 times.

▶ Iteration 1

▶ LSL r1, r1, #1: r1 = 0 << 1 = 0.

▶ AND r3, r0, #1: r3 = 0b1011_0010 & 1 = 0 (LSB of r0 is 0).

▶ ORR r1, r1, r3: r1 = 0 | 0 = 0.

▶ LSR r0, r0, #1: r0 = 0b0101_1001 (shift right, next LSB becomes current).

▶ SUB r2, r2, #1: r2 = 7.

▶ Effect: The first output bit appears at the LSB of r1 and equals original d0.

▶ Iteration 2

▶ LSL r1: r1 = 0 << 1 = 0.

▶ AND r3, r0, #1: r3 = 0b0101_1001 & 1 = 1 (new LSB is 1).

▶ ORR r1, r1, r3: r1 = 0 | 1 = 1 (now r1 = 0b0000_0001).

▶ LSR r0: r0 = 0b0010_1100.

▶ SUB r2: r2 = 6.

▶ Effect: r1 now holds bits “d0 d1”.

▶ On each subsequent iteration, r1 is shifted left before inserting the next bit, so earlier bits move toward the MSB while the newly inserted bit always starts at the LSB; over the whole loop, this builds the reversed word with the earliest extracted bits ending up toward the MSB by the end.

Test for Equal

- ▶ Give the different methods to test if two values held in registers `r0` and `r1` are equal.

Test for Equal ANS

- ▶ Give the different methods to test if two values held in registers r0 and r1 are equal.
- ▶ ANS:
 - ▶ ; Method 1: Compare instruction
 - ▶ `CMP r0, r1`
 - ▶ ; Method 2: Exclusive OR
 - ▶ `EORS r2, r0, r1` ; If equal, result is zero and Z flag set
 - ▶ ; Method 3: Subtract and test
 - ▶ `SUBS r2, r0, r1` ; If equal, result is zero and Z flag set

Summary: Condition Codes

Suffix	Description	Flags tested
EQ	E Qual	Z=1
NE	N ot E qual	Z=0
CS/HS	Unsigned H igher or S ame	C=1
CC/LO	Unsigned L ower	C=0
MI	M Inus (Negative)	N=1
PL	P Lus (Positive or Zero)	N=0
VS	o V erflow S et	V=1
VC	o V erflow C leared	V=0
HI	Unsigned H Igher	C=1 & Z=0
LS	Unsigned L ower or S ame	C=0 or Z=1
GE	Signed G reater or E qual	N=V
LT	Signed L ess T han	N!=V
GT	Signed G reater T han	Z=0 & N=V
LE	Signed L ess than or E qual	Z=1 or N!=V
AL	A Lways	

Note AL is the default and does not need to be specified

Conditional Instructions

- ▶ Write the following ARMv7 instructions:
 - ▶ Add registers r3 and r6 only if N is clear (from a previous instruction). Store the result in register r7.
 - ▶ Multiply registers r7 and r12, put the results in register r3 only if C is set and Z is clear (from a previous instruction)
 - ▶ Compare registers r6 and r8 only if Z is clear (from a previous instruction)

Conditional Instructions ANS

- ▶ Write the following ARMv7 instructions:
 - ▶ Add registers r3 and r6 only if N is clear. Store the result in register r7.
 - ▶ Multiply registers r7 and r12, put the results in register r3 only if C is set and Z is clear
 - ▶ Compare registers r6 and r8 only if Z is clear
- ▶ ANS:
 - ▶ ; Plus, Positive or Zero, (N=0). Add r3 and r6 only if N is clear, store in r7
 - ▶ ADDPL r7, r3, r6
 - ▶ ; HI, Unsigned Higher, (C=1 & Z=0). Multiply r7 and r12, put result in r3 only if C is set and Z is clear
 - ▶ MULHI r3, r7, r12
 - ▶ ; NE, Not equal (Z=0). Compare r6 and r8 only if Z is clear
 - ▶ CMPNE r6, r8

Assembly to C

- Write the equivalent C program for the following assembly code, assuming registers and C variables are related as (x=r0, y=r1). (Variables in C are in memory, and load/store assembly instructions are omitted here for brevity.) For example:

Assembly Program	C Program
CMP r0, #5 MOVEQ r0, #10 BLEQ fn	if (x == 5) { x = 10; fn(x); }
Assembly Program	C Program
CMP r0, #0 MOVLE r0, #0 MOVGT r0, #1	
Assembly Program	C rogram
CMP r0, #'A' CMPNE r0, #'B' MOVEQ r1, #1	

Assembly to C ANS

- Write the equivalent C program for the following assembly code, assuming registers and C variables are related as (x=r0, y=r1). (Variables in C are in memory, and load/store assembly instructions are omitted here for brevity.)

Assembly Program	C Program
CMP r0, #5 MOVEQ r0, #10 BLEQ fn	if (x == 5) { x = 10; fn(x); }
Assembly Program	C Program
CMP r0, #0 MOVLE r0, #0 MOVGT r0, #1	if (x <= 0) x = 0; else x = 1;
Assembly Program	C Program
CMP r0, #'A' CMPNE r0, #'B' MOVEQ r1, #1	if (c == 'A' c == 'B') y = 1;

Conditional Execution

- ▶ Rewrite the assembly program to use conditional execution statements.

Assembly Program	Assembly Program with Cond. Exec
CMP r3, #0 BEQ next ADD r0, r0, r1 SUB r0, r0, r2 next ...	

Conditional Execution ANS

- ▶ Rewrite the assembly program to use conditional execution statements.

Assembly Program	Assembly Program with Cond. Exec
CMP r3, #0 BEQ next ADD r0, r0, r1 SUB r0, r0, r2 next ...	CMP r3, #0 ADDNE r0, r0, r1 SUBNE r0, r0, r2 ...